

SYNERGY — Technical Specification

iOS Mobile Application · v1.0 · April 2026 · CONFIDENTIAL

This document covers the full technical architecture, iOS implementation requirements, API design, security model, data privacy framework, performance targets, and third-party service integrations required to build Synergy to production quality.

1. PLATFORM & ENVIRONMENT

Spec	Value	Notes
Primary platform	iOS 16.0+	Covers 94%+ of active iPhones as of 2026
Language	Swift 5.9 / SwiftUI	UIKit for complex custom animations only
Architecture	MVVM + Coordinator	Clean separation of view, logic, navigation
Min device	iPhone 11	A13 Bionic — required for on-device ML
Screen targets	390pt / 430pt / 375pt	iPhone 14 Pro / Max / SE — test all three
Orientation	Portrait only	Lock via Info.plist UISupportedInterfaceOrientations
Backend	Node.js 20 LTS on AWS	ECS Fargate — serverless container scaling
Database	PostgreSQL 15 (RDS)	PostGIS extension for geolocation matching
Cache layer	Redis 7 (ElastiCache)	Session tokens, match score cache, rate limiting
CDN	CloudFront + S3	User photos, chart SVGs, static assets

2. CORE FEATURE SPECIFICATIONS

2.1 Birth Chart Engine

The most technically critical component. All astrological calculations must be deterministic, auditable, and fast.

Calculation library

- Use Swiss Ephemeris (swisseph) via a dedicated Node.js microservice wrapped in a Swift Package Manager compatible REST client
- Calculate: Sun, Moon, Mercury, Venus, Mars, Jupiter, Saturn, Uranus, Neptune, Pluto, Ascendant (Rising), Midheaven, Chiron, North Node
- Require birth date, birth time (HH:MM), and birth city. Resolve city to lat/lng via Google Places API at capture time and store coordinates permanently
- House system: Placidus (most widely used in Western astrology). Expose as a user setting for advanced users — allow Koch or Whole Sign
- Store raw planetary positions (degrees) in database; never store derived scores. Recompute synastry scores on demand from raw data so scoring algorithm can be updated without data migration

Synastry calculation

- For each pair (User A, User B): compute all major aspects between A's planets and B's planets
- Aspect orbs: Conjunction $\pm 8^\circ$, Opposition $\pm 8^\circ$, Trine $\pm 6^\circ$, Square $\pm 6^\circ$, Sextile $\pm 4^\circ$, Quincunx $\pm 2^\circ$
- Weight aspects by planet significance: Venus $\times$ Venus (1.0), Moon $\times$ Moon (0.95), Mars $\times$ Venus (0.9), Sun $\times$ Sun (0.8), Rising $\times$ Rising (0.85)
- Cache synastry scores for 24 hours. Invalidate and recompute when transit modifier layer updates (daily at 00:01 UTC)

Transit modifier

- Daily job (AWS EventBridge) calculates current planetary positions at 00:01 UTC
- Score modifier: +/-15% cap on base synastry score based on active transits relevant to each user's chart
- Example: Venus transiting conjunct User A's natal Venus $\rightarrow$ +8% boost to all matches with strong Venus aspects

2.2 Matching Algorithm

Four weighted layers producing a 0-100 Cosmic Match Score™.

Spec	Value	Notes
Layer 1 — Synastry core	40% weight	Planet aspects, orb quality, mutual reception
Layer 2 — Elemental balance	25% weight	Fire/Water/Earth/Air distribution across both charts
Layer 3 — Intent alignment	20% weight	Relationship intention tags from onboarding
Layer 4 — Transit modifiers	15% weight	Daily planetary positions shift base score $\pm 15\%$

Feed algorithm

- Primary sort: Cosmic Match Score DESC
- Diversity injection: ensure no more than 3 consecutive profiles from same sun sign element
- Recency boost: profiles active within last 72h get +3 point boost to prevent cold-start for new users
- Mutual interest signal: if User B has already liked User A, boost A's position in B's feed by +12 points (not shown to user)
- Hard filters applied before scoring: location radius (user-set, default 25mi), age range, gender preference

2.3 Onboarding Flow

7 screens. Target 68% full completion. Each screen logs a funnel event to analytics.

Spec	Value	Notes
Screen 1 — Welcome	SwiftUI full-screen animation	SpriteKit star particle system. No auth required yet.
Screen 2 — Birth chart	UITextField + Google Places	Validate date with DateComponents. Store immediately on Next tap.
Screen 3 — Chart reveal	Custom animated SVG wheel	SVG rendered via WKWebView. Animate planets in sequence over 2.4s.
Screen 4 — Intention	Multi-select tag grid	Max 2 selections. Haptic feedback on select (UIImpactFeedbackGenerator).
Screen 5 — Profile	Photo picker + text fields	PHPickerViewController. Min 1 photo, max 6. Compress to 1200px max.
Screen 6 — Notifications	UNUserNotificationCenter	Request AFTER user taps Yes. Schedule 7am local time daily notification.
Screen 7 — First match	Match card reveal animation	Confetti + haptic on reveal. Log match_first_seen analytics event.

2.4 Real-Time Messaging

- WebSocket connection via Socket.io (server) / StarscreamSwift (iOS client)
- Message persistence: PostgreSQL with message_id UUID, sender_id, receiver_id, content (encrypted), created_at, read_at
- End-to-end encryption: Signal Protocol via libsignal-client. Keys generated on device, never transmitted to server
- Typing indicators: ephemeral socket events, never persisted
- Push notifications for new messages: APNs via AWS SNS. Payload contains match_id only — content fetched on open
- Message retention: 2 years for active matches, 90 days after unmatching (GDPR compliant)

2.5 Daily Horoscope & Ritual Engine

- AI generation: Claude API (claude-sonnet-4-6) called once per user per day via async queue
- Prompt includes: user's natal chart summary, current transits, user's match activity from last 48h
- Generated at 03:00 UTC, cached in Redis by user_id + date key, delivered on app open

- Ritual events: stored in events table with type (new_moon | full_moon | venus_season | retrograde), start_date, end_date, prompt_template
- New moon dates calculated 12 months ahead via Swiss Ephemeris and seeded into DB at deployment

3. API ARCHITECTURE

3.1 REST Endpoints — Core

Spec	Value	Notes
POST /auth/signup	Create account + chart	Returns JWT access_token (15min) + refresh_token (30d)
POST /auth/refresh	Refresh access token	Requires valid refresh_token in HttpOnly cookie
GET /users/me	Current user profile	Returns profile + natal chart summary
PUT /users/me	Update profile	Partial update. Recomputes chart if birth data changes.
GET /feed	Match feed	Paginated. Returns 20 matches with scores. ?cursor= for pagination
POST /feed/{id}/like	Like a profile	Returns {matched: bool}. Creates conversation if mutual.
POST /feed/{id}/pass	Pass on profile	Stores pass event. Profile hidden for 30 days.
GET /synastry/{user_id}	Compatibility report	Full synastry breakdown. Cached 24h. Requires Cosmic tier.
GET /horoscope/today	Daily horoscope	Returns AI-generated reading. Cached per user per day.
GET /conversations	All conversations	Sorted by last_message_at DESC. Unread count included.
GET /conversations/{id}/messages	Message history	Paginated. Cursor-based. Returns encrypted payloads.

3.2 Authentication & Sessions

- JWT access tokens (15-minute expiry) stored in iOS Keychain (kSecAttrAccessibleWhenUnlockedThisDeviceOnly)
- Refresh tokens (30-day expiry) sent as HttpOnly, Secure, SameSite=Strict cookie
- All API requests require Authorization: Bearer {token} header
- Rate limiting: 100 req/min per user via Redis sliding window. 429 response with Retry-After header
- Token rotation: refresh token rotated on each use. Previous refresh token invalidated immediately (refresh token reuse detection)
- Biometric auth (Face ID / Touch ID): LAContext.evaluatePolicy on app foreground. Protects Keychain access, not the token itself

4. DATA ARCHITECTURE & PRIVACY

4.1 Data Model — Key Tables

Spec	Value	Notes
users	id, email_hash, created_at, last_active, subscription_tier, location_point (PostGIS)	Email stored as bcrypt hash only
birth_charts	user_id, birth_date, birth_time, lat, lng, timezone, planetary_positions JSONB	positions = raw degrees, not derived
profiles	user_id, display_name, bio, intention_tags[], photos[] (S3 keys)	photos array stores S3 keys, not URLs

matches	id, user_a_id, user_b_id, cosmic_score, score_computed_at, status	status: liked matched unmatched
messages	id, conversation_id, sender_id, encrypted_content, iv, created_at, read_at	Content encrypted client-side
horoscopes	user_id, date, content, generated_at, model_version	Retained 90 days then purged
events	id, type, title, start_date, end_date, ritual_prompt	Cosmic calendar events
subscriptions	user_id, tier, stripe_subscription_id, current_period_end, cancelled_at	Tier: free cosmic oracle

4.2 GDPR & Privacy Compliance

- Data minimisation: only collect what's needed for core functionality. Birth time marked 'optional but recommended' in UI.
- Lawful basis: Contractual necessity (account) + Legitimate interests (matching algorithm) + Consent (marketing comms)
- Right to erasure: DELETE /users/me endpoint triggers: account deactivation → 30d grace period → hard delete of all PII. S3 photos deleted via lifecycle rule.
- Right to portability: GET /users/me/export returns JSON of all user data. Processed within 72 hours.
- Data residency: EU users stored in eu-west-1 (Ireland). UK users in eu-west-2 (London) post-Brexit. US users in us-east-1.
- Age verification: date of birth collected at signup. Block registration if under 18. Server-side validation — do not trust client.
- Privacy policy: plain English. Link on every auth screen. Version-controlled with effective date. Re-consent required on material changes.
- Data retention: active accounts retained indefinitely. Deleted accounts: PII purged in 30d, anonymised analytics retained.
- Location data: stored as PostGIS point (lat/lng). Approximate location shown to other users (nearest mile, not exact). Never share exact coordinates.

4.3 Security Requirements

- HTTPS only. HSTS header with 1-year max-age. Certificate pinning in iOS app (NSURLSession with TrustKit).
- All database connections use TLS 1.3. RDS encryption at rest (AES-256 via AWS KMS).
- User photos stored in private S3 bucket. Access via signed URLs (15-minute expiry). No public bucket access.
- API keys and secrets: AWS Secrets Manager. Never in source code or environment variables in plain text.
- Dependency scanning: Dependabot on GitHub. Block merges if critical CVE detected.
- Penetration testing: annual third-party pentest required before each major version release.
- Admin access: MFA required for all AWS console access. IAM roles with least-privilege principle.
- Abuse prevention: photo hash (pHash) checked against known CSAM databases via PhotoDNA API on upload.

5. IOS IMPLEMENTATION DETAIL

5.1 Project Structure

```

Synergy/
├── App/                # App entry, DI container, environment
├── Features/           # Feature modules (Onboarding, Feed, Chat, Profile...)
│   ├── Onboarding/    # ViewModels, Views, Coordinators
│   ├── Feed/
│   ├── Chat/
│   └── Today/          # Horoscope + ritual engine
├── Core/               # Networking, Auth, Analytics, Keychain
├── Services/           # AstrologyService, MatchService, NotificationService
├── Models/             # Codable data models
├── UI/                 # Design system — colours, typography, components
└── Resources/          # Assets.xcassets, Localizable.strings

```

5.2 Performance Targets

Spec	Value	Notes
App cold start	< 1.4 seconds	Measure via <i>MetricKit</i> . Baseline on iPhone 11.
Feed load (20 matches)	< 800ms	Includes image prefetch for first 5 cards
Chart reveal animation	60fps sustained	Use <i>CADisplayLink</i> . Avoid main thread work during animation.
Message send latency	< 200ms P95	<i>WebSocket</i> round-trip. Optimistic UI update immediately.
Horoscope load	< 300ms	Served from <i>Redis</i> cache. Fallback to DB if cache miss.
Photo upload	< 3s for 5MB	Compress to 1200px max before upload. Show progress.
Memory ceiling	< 150MB typical	Profile with <i>Instruments</i> . Leak detection in CI.
Battery impact	< 2% per hour active use	No background location. Minimal background refresh.
Crash-free rate	> 99.5%	Monitor via <i>Firebase Crashlytics</i> . Alert if drops below.

5.3 Key iOS Frameworks & Dependencies

Spec	Value	Notes
SwiftUI	Primary UI framework	iOS 16+ for <i>NavigationStack</i> , <i>.searchable</i> , etc.
Combine	Reactive data flow	<i>Publishers</i> from <i>ViewModels</i> to <i>Views</i>
URLSession	HTTP networking	<i>Async/await</i> . Custom retry logic for 5xx errors.
Starscream	WebSocket (SPM)	github.com/daltoniam/Starscream — chat connections
SDWebImageSwiftUI	Image loading/caching (SPM)	Async image load with placeholder + fade
WKWebView	Chart SVG rendering	Sandboxed. No JS bridge for security. Message handler only.
SpriteKit	Star particle system	Welcome screen animation. Preload on launch.
PhotosUI (PHPickerViewController)	Photo selection	Privacy-respecting. No full library access required.
UserNotifications	Push + local notifications	APNs for messages. Local for daily horoscope reminder.
LocalAuthentication	Face ID / Touch ID	Protect <i>Keychain</i> access on foreground.
CoreLocation (minimal)	City resolution only	<i>CLGeocoder</i> for onboarding. Not used after. No background.
StoreKit 2	Subscriptions	In-app purchase for <i>Cosmic/Oracle</i> . Server-side receipt validation.
TrustKit	Certificate pinning (CocoaPods)	Pin to Synergy API cert. Alert on pin failure.

5.4 Accessibility

- VoiceOver: all interactive elements must have `accessibilityLabel` and `accessibilityHint`
- Dynamic Type: all text must scale with `UIContentSizeCategory`. Test with largest accessibility size.
- Minimum tap target: 44×44pt for all interactive elements (Apple HIG requirement)
- Colour contrast: minimum 4.5:1 for body text, 3:1 for large text (WCAG AA)

- Reduce Motion: respect `UIAccessibility.isReduceMotionEnabled`. Replace animations with cross-fades.
- Dark mode: full support required. Test all screens in both modes before release.

6. THIRD-PARTY SERVICE INTEGRATIONS

Spec	Value	Notes
Swiss Ephemeris	Astrological calculations	LGPL licence. Run as internal microservice — do not expose directly.
Google Places API	City autocomplete + geocoding	Enable Places (New) API. Cache city → lat/lng to reduce API calls.
Stripe	Subscription billing	Use Stripe Billing + StoreKit 2 for iOS IAP. Webhook for subscription events.
Claude API (Anthropic)	Horoscope generation	claude-sonnet-4-6. Rate limit: 1 call per user per day. Cache output.
APNs (Apple)	Push notifications	Use HTTP/2 APNs API directly. Token-based auth (p8 key).
AWS S3	Photo storage	Private bucket. Presigned URLs for upload + delivery.
AWS ElastiCache (Redis)	Caching + session store	Cluster mode enabled. AOF persistence for session data.
Firebase Crashlytics	Crash monitoring	iOS SDK. No analytics. Crash-free rate dashboard alert.
Mixpanel	Product analytics	Funnel events only. No PII in event properties. Pseudonymised user_id.
PhotoDNA (Microsoft)	CSAM detection	API call on every photo upload. Block + report if match found.
Twilio SendGrid	Transactional email	Password reset, subscription receipts. Unsubscribe required.

7. TESTING STRATEGY

7.1 iOS Testing

- Unit tests: XCTest. Target 80%+ coverage for ViewModels, Services, and Models
- UI tests: XCUITest for critical flows — onboarding, match like, message send, subscription purchase
- Snapshot tests: `iOSSnapshotTestCase` for design-sensitive components (match card, chart wheel, tier cards)
- Performance tests: `XCTMetric` for chart calculation time, feed load time. Fail build if regression > 20%
- TestFlight: internal (team) → external (beta group of 500) → App Store. 2-week beta minimum before release

7.2 Backend Testing

- Unit tests: Jest. 90%+ coverage on astrological calculation service (must be deterministic)
- Integration tests: Supertest against local DB. Full API contract tests.
- Load testing: k6. Target: 1,000 concurrent users, 95th percentile latency < 500ms
- Security testing: OWASP ZAP automated scan in CI. Fail if critical or high findings.

8. DEPLOYMENT & CI/CD

Spec	Value	Notes
iOS CI	Xcode Cloud	Build → Test → Archive → TestFlight on every PR merge to main
Backend CI	GitHub Actions	Lint → Test → Docker build → Deploy to staging on merge

Staging environment	AWS ECS Fargate (eu-west-2)	Mirror of prod. Used for QA and TestFlight builds.
Production deployment	Blue/green via AWS CodeDeploy	Zero-downtime deploys. Automatic rollback on health check failure.
Database migrations	Flyway	Migrations run automatically on deploy. Rollback script required for each migration.
Feature flags	LaunchDarkly	All new features behind flags. Gradual rollout — 5% → 25% → 100%
Monitoring	AWS CloudWatch + Datadog	API latency, error rate, DB connections. PagerDuty alerts for P1 issues.
App Store	Manual approval step	Two-person sign-off required before App Store submission.

9. RISK REGISTER

ID	Risk	Impact	Likelihood	Mitigation
R1	App Store rejection for dating app features	High	Low	Follow App Store Review Guidelines 1.1.4. No adult content. Age gate at 18. Submit with detailed review notes.
R2	Astrology calculation inaccuracy damages trust	High	Medium	Swiss Ephemeris is industry standard. Add disclaimer that app is for entertainment/connection. User testing with astrology experts pre-launch.
R3	Data breach / PII exposure	High	Low	E2E encryption on messages. AES-256 at rest. Annual pentest. Incident response plan documented. Cyber insurance required.
R4	AI horoscope content quality issues	Medium	Medium	Human review of 50 samples per week in first 3 months. Moderation pipeline for flagged content. Rate limit generations.
R5	Cold-start matching (not enough users in area)	High	High	Launch in London only. Geofence expansion tied to density thresholds (min 500 users within 25mi before expanding city). Show 'expanding soon' otherwise.
R6	StoreKit / Stripe billing disputes	Medium	Medium	Server-side receipt validation. Grace period on lapsed subscriptions. Clear cancellation flow. Monitor chargeback rate.
R7	Copycat product from incumbent (Hinge, Bumble)	Medium	High	Move fast on community and brand. Astrology community is loyal. Patent provisional on Cosmic Match Score algorithm. Build switching costs via match history.

10. GLOSSARY

Synastry: The astrological practice of comparing two birth charts to assess compatibility by analysing aspects between planetary positions.

Natal chart: A map of the sky at the exact moment and location of a person's birth, showing planetary positions in the zodiac.

Transit: The current real-time position of a planet in relation to a natal chart position.

Aspect: An angular relationship between two planets (e.g. conjunction = 0°, trine = 120°, opposition = 180°).

Orb: The allowable deviation from an exact aspect angle. A 6° orb on a trine means 114°–126° counts.

Placidus: A house system dividing the sky into 12 sections. Most common in Western astrology.

Swiss Ephemeris: High-precision astronomical calculation library used as the industry standard for astrological software.

Cosmic Match Score™: Synergy's proprietary 0-100 compatibility score combining synastry, elemental balance, intent, and transits.

JWT: JSON Web Token — a signed token used for stateless authentication between client and server.

PostGIS: PostgreSQL extension enabling geographic object storage and spatial queries (used for location-based matching).